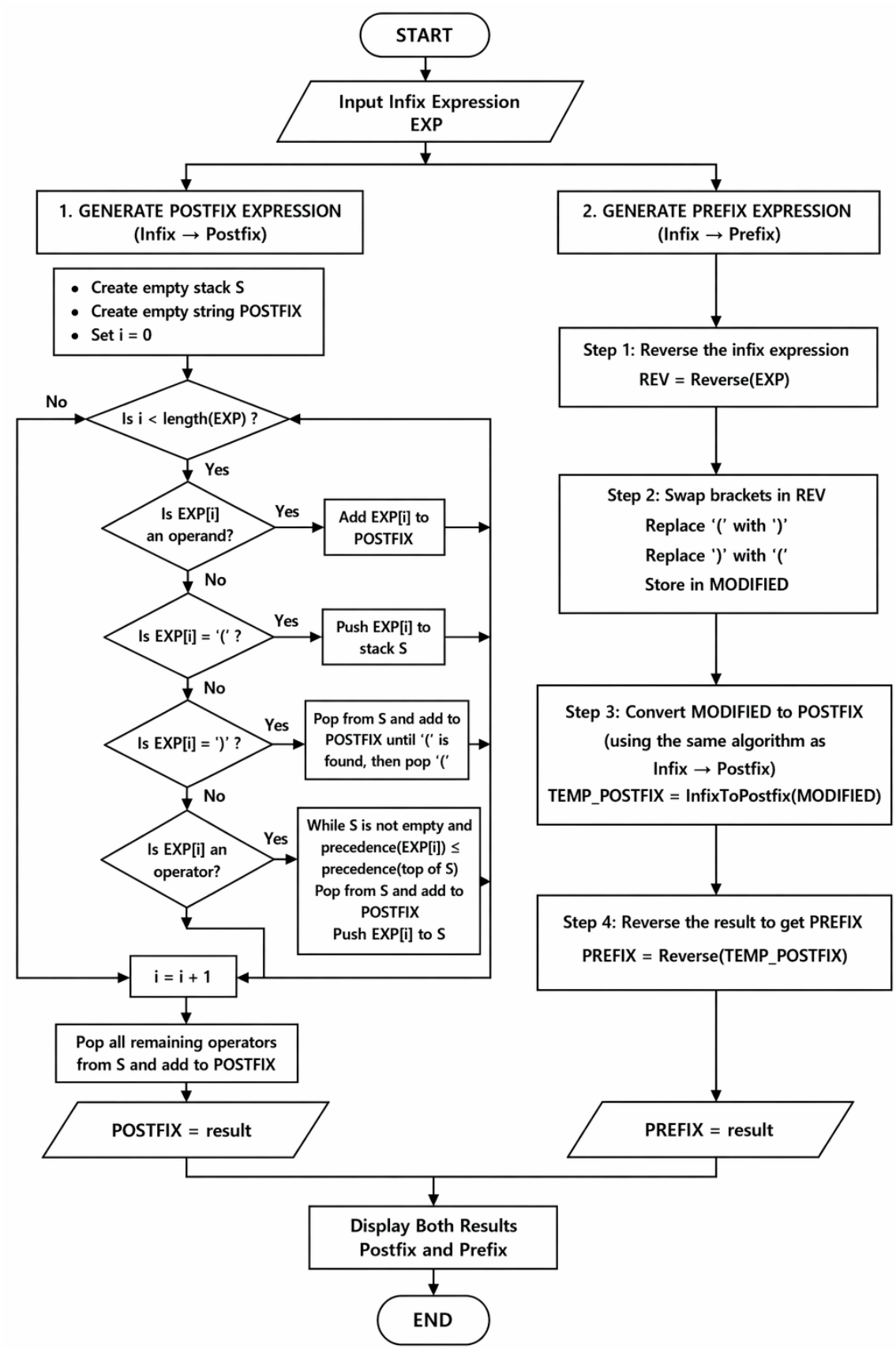
START
Input Infix Expression
EXP
1. GENERATE POSTFIX EXPRESSION
(Infix → Postfix)
2. GENERATE PREFIX EXPRESSION
(Infix → Prefix)
Create empty stack S
Create empty string POSTFIX
Set i = 0
Is i < length(EXP) ?
No
Yes
Is EXP[i]
an operand?
Yes
Add EXP[i] to
POSTFIX
No
Is EXP[i] = '(' ?
Yes
Push EXP[i] to
stack S
No
Is EXP[i] = ')' ?
Yes
Pop from S and add to
POSTFIX until '(' is
found, then pop '('
No
Is EXP[i] an
operator?
Yes
While S is not empty and
precedence(EXP[i]) ≤
precedence(top of S)
Pop from S and add to
POSTFIX
Push EXP[i] to S
i = i + 1
Pop all remaining operators
from S and add to POSTFIX
POSTFIX = result
Step 1: Reverse the infix expression
REV = Reverse(EXP)
Step 2: Swap brackets in REV
Replace '(' with ')'
Replace ')' with '('
Store in MODIFIED
Step 3: Convert MODIFIED to POSTFIX
(using the same algorithm as
Infix → Postfix)
TEMP_POSTFIX = InfixToPostfix(MODIFIED)
Step 4: Reverse the result to get PREFIX
PREFIX = Reverse(TEMP_POSTFIX)
PREFIX = result
Display Both Results
Postfix and Prefix
END